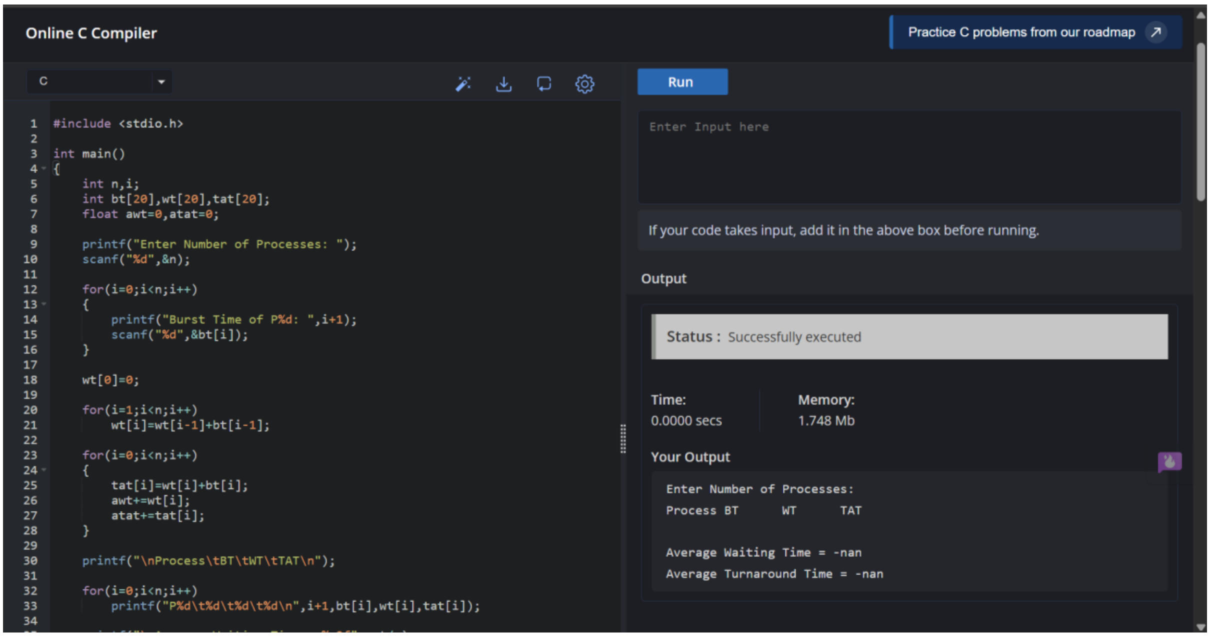
EXPERIMENT 03:

4. Construct a scheduling program with C that selects the waiting process with the smallest execution time to execute
next.



PSEUDO CODE :

Start

Input number of processes

Input Burst Time

Sort processes by Burst Time

Waiting Time of first process = 0

For remaining processes

Waiting Time = Sum of previous burst times

Turnaround Time = Waiting Time + Burst Time

Calculate averages

Display results

Stop

Process	BT	WT	TAT
P3	2	0	2
P2	3	2	5
P1	5	5	10

Average Waiting Time = 2.33

Average Turnaround Time = 5.67

CODE:

```
#include <stdio.h>

int main()
{
   int n,i,j,temp;
   int bt[20],wt[20],tat[20];

   float awt=0,atat=0;

   printf("Enter Number of Processes: ");
   scanf("%d",&n);

   for(i=0;i<n;i++)
   {
      printf("Burst Time of P%d: ",i+1);
```

```
   scanf("%d",&bt[i]);
}

for(i=0;i<n;i++)
{
   for(j=i+1;j<n;j++)
   {
      if(bt[i]>bt[j])
      {
         temp=bt[i];
         bt[i]=bt[j];
         bt[j]=temp;
      }
   }
}

wt[0]=0;

for(i=1;i<n;i++)
   wt[i]=wt[i-1]+bt[i-1];

for(i=0;i<n;i++)
{
   tat[i]=wt[i]+bt[i];
   awt+=wt[i];
   atat+=tat[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for(i=0;i<n;i++)
   printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);

printf("\nAverage Waiting Time = %.2f",awt/n);
```

```c
    printf("\nAverage Turnaround Time = %.2f",atat/n);


    return 0;
}
```